

Kushal Jain

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Mumbai, Maharashtra

SUMMARY

Final-year B.Tech. student in Computer Science (AI & Data Science) at MIT World Peace University with hands-on experience building production-ready ML systems, GenAI applications, and full-stack solutions. Track record of delivering measurable results through hackathon wins and deployed projects spanning computer vision, predictive analytics, and LLM-powered applications.

AWARDS AND ACHIEVEMENTS

- **Winner – BITS Pilani National Hackathon:** Secured first place among 100+ teams by delivering production-ready end-to-end solution prototype in 36 hours
- **Student Achiever Award – MIT World Peace University:** Recognized for outstanding academic performance, technical contributions, and leadership among 800+ graduating students
- **Team Lead & Finalist – Pune MahaMetro Hackathon:** Led 6-member team delivering 4 CV safety prototypes for metro operations achieving 92% mAP on 15,000+ frames with 30 FPS processing

PROJECTS

- **AI Tutor: Voice-Interactive Learning Platform** *Jan 2024 – Present*
 - Designed interactive AI tutor with multimodal capabilities including speech-to-text, text-to-speech, and whiteboard visualization, improving learner engagement by 40%
 - Engineered Neo4j knowledge graph with 50+ concept nodes enabling adaptive difficulty and personalized learning paths based on individual performance patterns
 - Integrated LangChain for structured LLM responses, organizing content into concepts
 - **Tech Stack:** Python, Flask, React, Neo4j, VAPI, ElevenLabs, LangChain
- **Workload Spike Prediction & Auto-Scaling System** *Jan 2025 – Feb 2025*
 - Debugged and optimized batch prediction pipeline combining Prophet time-series forecasting with LLM-based anomaly detection, reducing false-positives by 35%
 - Deployed containerized solution with Docker and real-time monitoring dashboards in Grafana for production visibility
 - **Tech Stack:** Python, Kubernetes, Prophet, Docker, Grafana, LLMs
- **Real-Time Parking Occupancy Detection System** *Sep 2024 – Dec 2024*
 - Built computer vision system achieving 94% detection accuracy for real-time parking slot monitoring across varied lighting and weather conditions
 - Developed modular architecture supporting multiple camera feeds with centralized occupancy dashboard for urban parking management
 - **Tech Stack:** Python, OpenCV, YOLOv5, Docker

SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript **Backend & DevOps:** Flask, FastAPI, Docker, Kubernetes, Git, CI/CD, REST APIs
- **ML & AI:** PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, Transformers, LangChain, RAG, LLM Fine-tuning
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Hypothesis Testing, A/B Testing, EDA, Feature Engineering
- **Databases:** MySQL, MongoDB, Neo4j, Hadoop, Query Optimization **Tools:** Power BI, Tableau, Jupyter, GCP, AWS, Grafana

EDUCATION

- **MIT World Peace University** Pune, Maharashtra
 - B.Tech. in Computer Science (Artificial Intelligence & Data Science); CGPA: 7.2/10* *Jun 2022 – Jul 2026*
- **Shree L.R. Tiwari College of Science and Commerce** Mumbai, Maharashtra
 - Class XII (HSC Board); Percentage: 76%* *Mar 2022*